

1. GENERATE RAW MOCK DATA

```
import pandas as pd
import numpy as np

raw_data = {
    "Employee_Name": [" Alice ", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, np.nan, 92000, 60000],
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28"
    ]
}

df = pd.DataFrame(raw_data)

print("--- Raw Data ---")
print(df)
```

```
... --- Raw Data ---
   Employee_Name  Department  Salary  Join_Date
0         Alice           HR  50000.0  2022-01-15
1          Bob    Engineering  85000.0  2021-06-20
2      Charlie           HR      NaN  2023-03-11
3        Diana    Engineering  92000.0  2020-11-01
4          Evan     Marketing  60000.0  2024-02-28
```

2. DATA PREPROCESSING

```
df["Employee_Name"] = df["Employee_Name"].str.strip()

median_salary = df["Salary"].median()
df["Salary"] = df["Salary"].fillna(median_salary)

df["Join_Date"] = pd.to_datetime(df["Join_Date"])

df["Years_of_Service"] = 2026 - df["Join_Date"].dt.year

print("--- Preprocessed Data ---")
print(df)
```

```
... --- Preprocessed Data ---
   Employee_Name  Department  Salary  Join_Date  Years_of_Service
0         Alice           HR  50000.0  2022-01-15                4
1          Bob    Engineering  85000.0  2021-06-20                5
2      Charlie           HR  72500.0  2023-03-11                3
3        Diana    Engineering  92000.0  2020-11-01                6
4          Evan     Marketing  60000.0  2024-02-28                2
```

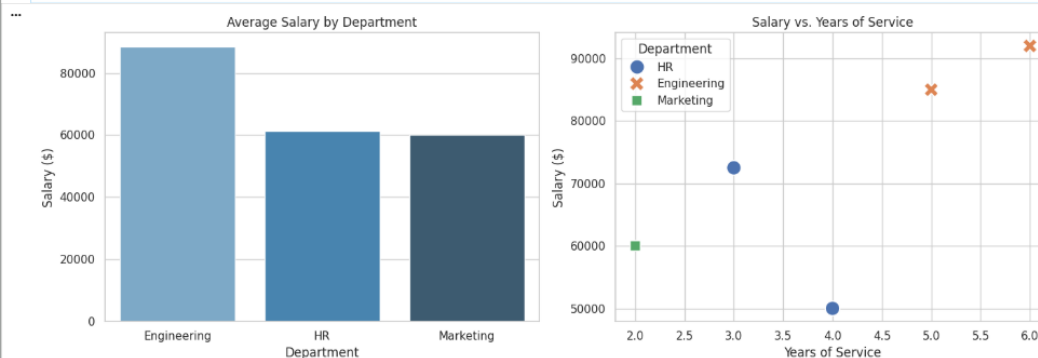
3. DATA AGGREGATION

```
dept_summary = (  
    df.groupby("Department")  
        .agg(  
            Avg_Salary=("Salary", "mean"),  
            Total_Employees=("Employee_Name", "count")  
        )  
        .reset_index()  
    )  
  
print("--- Aggregated Summary ---")  
print(dept_summary)
```

```
... --- Aggregated Summary ---  
      Department  Avg_Salary  Total_Employees  
0  Engineering      88500.0             2  
1           HR      61250.0             2  
2    Marketing      60000.0             1
```

4. DATA VISUALIZATION

```
import matplotlib.pyplot as plt  
import seaborn as sns  
  
sns.set_theme(style="whitegrid")  
  
fig, axes = plt.subplots(1, 2, figsize=(14, 5))  
  
sns.barplot(  
    data=dept_summary,  
    x="Department",  
    y="Avg_Salary",  
    hue="Department",  
    palette="Blues_d",  
    legend=False,  
    ax=axes[0]  
)  
  
axes[0].set_title("Average Salary by Department")  
axes[0].set_xlabel("Department")  
axes[0].set_ylabel("Salary ($)")  
  
sns.scatterplot(  
    data=df,  
    x="Years_of_Service",  
    y="Salary",  
    hue="Department",  
    style="Department",  
    s=200,  
    ax=axes[1]  
)  
  
axes[1].set_title("Salary vs. Years of Service")  
axes[1].set_xlabel("Years of Service")  
axes[1].set_ylabel("Salary ($)")  
  
plt.tight_layout()  
plt.show()
```



5. Creating Sample DataFrame

```
▶ import pandas as pd

df = pd.DataFrame()

df["Name"] = ["John", "Emma", "Liam", "Olivia"]
df["Age"] = [20, 19, 21, 18]
df["Student"] = [True, True, False, True]

print(df)
```

```
...      Name  Age  Student
0     John   20     True
1     Emma   19     True
2     Liam   21    False
3  Olivia   18     True
```

6. Adding Row to DataFrame

```
▶ new_row = pd.DataFrame(
    [
        ["Sophia", 22, False],
    ],
    columns=["Name", "Age", "Student"]
)

df = pd.concat([df, new_row], ignore_index=True)

print(df)
```

```
...      Name  Age  Student
0     John   20     True
1     Emma   19     True
2     Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
```

7. Measures of Central Tendency Using SciPy

```
▶ from scipy import stats
import numpy as np

data = [10, 20, 30, 40, 50]

print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data))

... Mean: 30.0
    Median: 30.0
    Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
```

8. Probability Distribution Analysis Using SciPy

```
▶ from scipy.stats import norm

probability = norm.cdf(85, loc=70, scale=10)

print("Probability:", round(probability, 4))

... Probability: 0.9332
```

9. Hypothesis Testing

```
▶ from scipy import stats

data = [22, 25, 19, 24, 28, 30]

t_stat, p_value = stats.ttest_1samp(data, 25)

print("T-Statistic:", round(t_stat, 3))
print("P-Value:", round(p_value, 3))

... T-Statistic: -0.205
    P-Value: 0.846
```

10. Correlation Analysis

```
▶ from scipy.stats import pearsonr

x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

corr, p_value = pearsonr(x, y)

print("Correlation:", corr)
```

```
... Correlation: 1.0
```

11. Linear Algebra Operations

```
▶ from scipy import linalg

A = [[3, 2],
     [1, 2]]

B = [5, 5]

solution = linalg.solve(A, B)

print(solution)
```

```
... [0.  2.5]
```

12. Optimization Using SciPy

```
▶ from scipy.optimize import minimize

def objective(x):
    return x**2 + 4

result = minimize(objective, x0=5)

print(result.x)
```

```
... [-2.62955131e-08]
```